

CI/CD Pipeline Documentation

1.Aim

Configure a complete Continuous Integration and Continuous Deployment (CI/CD) pipeline using:

- Red Hat Enterprise Linux 9.8
- Jenkins
- Apache Maven
- Git & GitHub
- Apache Tomcat

The pipeline automatically builds a Java Maven Web Application from GitHub and deploys it to the Apache Tomcat server.

2. Prerequisites

Before starting, ensure:

- Two RHEL 9.8 Virtual Machines
 - Jenkins Server
 - Tomcat Server
- Internet connectivity
- Root privileges
- Mounted RHEL DVD repository
- Firewall service enabled

3. Jenkins Server Configuration

Step 1- Verify Network Connectivity

Configure the virtual machine to use the NAT network adapter and verify internet connectivity using the following commands:

```
#ping www.google.com  
#ping www.redhat.com
```

```
[root@jenkins ~]#  
[root@jenkins ~]# ping www.google.com  
PING www.google.com (142.251.153.119) 56(84) bytes of data.  
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=1 ttl=128 time=30.4 ms  
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=2 ttl=128 time=28.2 ms  
^C  
--- www.google.com ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1001ms  
rtt min/avg/max/mdev = 28.157/29.298/30.440/1.141 ms  
[root@jenkins ~]# ping www.redhat.com  
PING e3396.dscx.akamaiedge.net (23.63.221.112) 56(84) bytes of data.  
64 bytes from a23-63-221-112.deploy.static.akamaitechnologies.com (23.63.221.112): icmp_seq=1 t  
tl=128 time=33.6 ms  
64 bytes from a23-63-221-112.deploy.static.akamaitechnologies.com (23.63.221.112): icmp_seq=2 t  
tl=128 time=43.6 ms  
^C  
--- e3396.dscx.akamaiedge.net ping statistics ---  
3 packets transmitted, 2 received, 33.333% packet loss, time 2004ms  
rtt min/avg/max/mdev = 33.638/38.631/43.624/4.993 ms
```

Step 2- Take VM Snapshot

Step 3- Configure Hostname

```
#hostnamectl set-hostname jenkins.example.com
```

```
[root@jenkins ~]#  
[root@jenkins ~]# hostnamectl set-hostname jenkins.example.com  
[root@jenkins ~]# hostname  
jenkins.example.com
```

Step 4- Mount RHEL DVD

```
#mount /dev/cdrom /mnt
```

```
#yum clean all
```

```
#yum repolist all
```

```
[root@jenkins ~]# mount /dev/cdrom /mnt
mount: /mnt: WARNING: source write-protected, mounted read-only.
[root@jenkins ~]# yum clean all
Updating Subscription Management repositories.

This system is registered with an entitlement server, but is not receiving updates. You can use
subscription-manager to assign subscriptions.


36 files removed
[root@jenkins ~]# yum repolist all
Updating Subscription Management repositories.

This system is registered with an entitlement server, but is not receiving updates. You can use
subscription-manager to assign subscriptions.


repo id                                repo name                               status
AppStream                              this is cdrom repo of ap enabled
BaseOS                                  This is a baseos repo                  enabled
```

Step 5- Install Java 21 and Verify installation

Java is required because both Jenkins and Maven run on Java. Install Java using the following command:

```
#yum install java-21-openjdk -y
```

```
#java --version
```

```
Complete!  
[root@jenkins ~]# java --version  
openjdk 21.0.11 2026-04-21 LTS  
OpenJDK Runtime Environment (Red_Hat-21.0.11.0.10-1) (build 21.0.11+10-LTS)  
OpenJDK 64-Bit Server VM (Red_Hat-21.0.11.0.10-1) (build 21.0.11+10-LTS, mixed mode, sharing)
```

Step 6- Apache Maven Installation

Download Maven

```
#cd /opt
```

```
#wget https://dlcdn.apache.org/maven/maven-3/3.9.16/binaries/apache-maven-3.9.16-bin.tar.gz
```

```
[root@jenkins ~]# cd /opt  
[root@jenkins opt]# wget https://dlcdn.apache.org/maven/maven-3/3.9.16/binaries/apache-maven-3.9.16-bin.tar.gz  
--2026-07-02 22:26:41-- https://dlcdn.apache.org/maven/maven-3/3.9.16/binaries/apache-maven-3.9.16-bin.tar.gz  
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644  
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 9278065 (8.8M) [application/x-gzip]  
Saving to: 'apache-maven-3.9.16-bin.tar.gz'  
  
apache-maven-3.9.16-bin. 100%[=====>] 8.85M 362KB/s in 13s  
2026-07-02 22:26:57 (709 KB/s) - 'apache-maven-3.9.16-bin.tar.gz' saved [9278065/9278065]
```

Step 7- Extract and rename

```
#tar -xvzf apache-maven-3.9.16-bin.tar.gz
```

```
#mv apache-maven-3.9.16 maven
```

```
[root@jenkins opt]# tar -xvzf apache-maven-3.9.16-bin.tar.gz  
apache-maven-3.9.16/README.txt  
apache-maven-3.9.16/LICENSE  
apache-maven-3.9.16/NOTICE  
apache-maven-3.9.16/lib/  
apache-maven-3.9.16/lib/aopalliance.license  
apache-maven-3.9.16/lib/asm.license  
apache-maven-3.9.16/lib/commons-cli.license
```

Step 8- Configure Environment Variables

Edit ~/.bashrc

Add the following environment variables to configure Java and Maven paths-

JAVA_HOME=/usr/lib/jvm/java-21-openjdk

M2_HOME=/opt/maven

M2=/opt/maven/bin

PATH=\$PATH:\$HOME/bin:\$JAVA_HOME:\$M2:\$M2_HOME

export PATH

```
alias rm='rm -i'
alias cp='cp -i'
alias mv='mv -i'

#####for configure path of maven#####
JAVA_HOME=/usr/lib/jvm/java-21-openjdk

M2_HOME=/opt/maven

M2=/opt/maven/bin

PATH=$PATH:$HOME/bin:$JAVA_HOME:$M2:$M2_HOME

export PATH
-- INSERT --
```

Reload the .bashrc file so the newly added environment variables take effect.

source ~/.bashrc

```
[root@jenkins ~]#
[root@jenkins ~]# source .bashrc
[root@jenkins ~]# mvn -version
Apache Maven 3.9.16 (2bdd9fddda4b155ebf8000e807eb73fd829a51d5)
Maven home: /opt/maven
Java version: 17.0.19, vendor: Red Hat, Inc., runtime: /usr/lib/jvm/java-17-openjdk-17.0.19.0.10-2.el9.x86_64
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "5.14.0-687.17.1.el9_8.x86_64", arch: "amd64", family: "unix"
[root@jenkins ~]#
```

Step 9- Create Maven Web Application

mvn archetype:generate | grep maven-archetype-webapp

mvn archetype:generate

```
[root@jenkins ~]# mvn archetype:generate | grep maven-archetype-webapp
740: remote -> com.haoxuer.maven.archetype:maven-archetype-webapp (a simple maven archetype)
903: remote -> com.lodsve:lodsve-maven-archetype-webapp (Lodsve Maven Archetype Webapp)
2362: remote -> org.apache.maven.archetypes:maven-archetype-webapp (An archetype which contains a sample Maven Webapp project.)
2630: remote -> org.bytesizebook.com.guide.boot:maven-archetype-webapp (An archetype to create the starting code for the first three chapters of Guide to Web Development with Java, 2nd edition.)
^C
[root@jenkins ~]# mvn archetype:generate
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Interactive mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
Choose archetype:
```

```

3681: remote -> za.co.absa.hyperdrive:component-archetype (-)
3682: remote -> za.co.absa.hyperdrive:component-archetype_2.11 (-)
3683: remote -> za.co.absa.hyperdrive:component-archetype_2.12 (-)
Choose a number or apply filter (format: [groupId:]artifactId, case sensitive contains): 2357: 2362
Choose org.apache.maven.archetypes:maven-archetype-webapp version:
1: 1.0-alpha-1
2: 1.0-alpha-2
3: 1.0-alpha-3
4: 1.0-alpha-4
5: 1.0
6: 1.3
7: 1.4
8: 1.5
Choose a number: 8: 8
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.5/maven-archetype-webapp-1.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.5/maven-archetype-webapp-1.5.jar (9.9 kB at 9.9 kB/s)
Define value for property 'groupId': in.radical.webapp
Define value for property 'artifactId': radical-webapp
Define value for property 'version' 1.0-SNAPSHOT:
Define value for property 'package' in.radical.webapp:
Confirm properties configuration:
groupId: in.radical.webapp
artifactId: radical-webapp
version: 1.0-SNAPSHOT
package: in.radical.webapp
Y: Y

```

The Maven Archetype generates the initial source code and project structure for the web application. This generated project will be used throughout the CI/CD pipeline. Navigate to the generated project directory (radical-webapp) and continue with the remaining steps.

After the project is created, verify that the directory exists. Run:

```
#ls
```

Navigate into the project directory.

```
#cd radical-webapp
```

Display the project structure.

```
#tree .
```

```

[ERROR] [http-1] http://wiki.apache.org/confluence/display/MAVEN/Installing+Project+Exception
[root@jenkins ~]# pwd
/root
[root@jenkins ~]# ls
adduser.yml      Desktop  Downloads  Pictures  radical-webapp  Videos
anaconda-ks.cfg  Documents Music      Public    Templates
[root@jenkins ~]# cd radical-webapp/
[root@jenkins radical-webapp]# ll
total 4
-rw-r--r--. 1 root root 2220 Jul  2 22:52 pom.xml
drwxr-xr-x. 3 root root  18 Jul  2 22:52 src
[root@jenkins radical-webapp]# tree .
.
├── pom.xml
├── src
│   └── main
│       └── webapp
│           ├── index.jsp
│           ├── WEB-INF
│           └── web.xml
4 directories, 3 files
[root@jenkins radical-webapp]#

```

Run the Maven lifecycle commands one by one to verify that the project builds successfully.

- `#mvn validate` - Validates the project structure.
- `#mvn compile` - Compiles the source code.
- `#mvn test` - Executes unit tests.
- `#mvn package` - Packages the application and generates the WAR file.

```
cd radical-webapp/  
ll  
tree .  
mvn validate  
mvn test  
mvn compile  
mvn test  
mvn package  
tree .  
history
```

```
137 history  
[root@jenkins radical-webapp]# tree .  
.  
├── pom.xml  
├── src  
│   └── main  
│       └── webapp  
│           ├── index.jsp  
│           └── WEB-INF  
│               └── web.xml  
└── target  
    ├── maven-archiver  
    │   └── pom.properties  
    ├── radical-webapp  
    │   ├── index.jsp  
    │   ├── META-INF  
    │   ├── WEB-INF  
    │   │   └── classes  
    │   └── web.xml  
    └── radical-webapp.war  
  
10 directories, 7 files  
[root@jenkins radical-webapp]#
```

Step 10- Git & GitHub Configuration

Install Git

```
# yum install git -y
```

```
Installing...
Installed products updated.
```

```
Downgraded:
```

```
git-core-2.39.1-1.el9.x86_64
```

```
Installed:
```

```
git-2.39.1-1.el9.x86_64
```

```
perl-Error-1:0.17029-7.el9.noarch
```

```
perl-TermReadKey-2.38-11.el9.x86_64
```

```
git-core-doc-2.39.1-1.el9.noarch
```

```
perl-Git-2.39.1-1.el9.noarch
```

```
perl-lib-0.65-480.el9.x86_64
```

```
Complete!
```

```
[root@jenkins ~]# git --version
```

```
git version 2.39.1
```

```
[root@jenkins ~]#
```

Initialize the git repository

```
#git init .
```

```
[root@jenkins radical-webapp]# git init .
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /root/radical-webapp/.git/
```

Configure Git by adding your username and email.

```
#git config --global user.name "andy"
```

```
#git config --global user.email "andy@example.com"
```

```
root@jenkins:~/radical-webapp
[rootejenkins radical-webapp]# git config --global user.name "andy"
[rootejenkins radical-webapp]#
[rootejenkins radical-webapp]# git config --global user.email "andy@example.com"
[rootejenkins radical-webapp]#
```

```
[root@jenkins radical-webapp]# git config --global --list
user.name=andy
user.email=andy@example.com
[root@jenkins radical-webapp]#
```

Stage all files so they are ready to be committed.

git add .

```
[root@jenkins radical-webapp]# git add .
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]# git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   .mvn/jvm.config
    new file:   .mvn/maven.config
    new file:   pom.xml
    new file:   src/main/webapp/WEB-INF/web.xml
    new file:   src/main/webapp/index.jsp
    new file:   target/maven-archiver/pom.properties
    new file:   target/radical-webapp.war
    new file:   target/radical-webapp/WEB-INF/web.xml
    new file:   target/radical-webapp/index.jsp
```

Once all required changes have been staged, create a commit to save the current version of the project.

git commit -m "Initial Commit" #git commit -m "code1 my originalcode"

```
[root@jenkins radical-webapp]# git commit -m "code1 my originalcode"
[master (root-commit) d68b7d3] code1 my originalcode
 9 files changed, 100 insertions(+)
 create mode 100644 .mvn/jvm.config
 create mode 100644 .mvn/maven.config
 create mode 100644 pom.xml
 create mode 100644 src/main/webapp/WEB-INF/web.xml
 create mode 100644 src/main/webapp/index.jsp
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/radical-webapp.war
 create mode 100644 target/radical-webapp/WEB-INF/web.xml
 create mode 100644 target/radical-webapp/index.jsp
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]#
```

To verify we can see in git log

#git log

```
[root@jenkins radical-webapp]# git log
commit d68b7d32e9b312a16f456ce644f39234e11ac5a6 (HEAD -> master)
Author: andyy <andy@example.com>
Date:   Tue Jul 7 21:12:47 2026 +0530

    code1 my originalcode
[root@jenkins radical-webapp]# █
```

Connect GitHub repository

Create a new repository in your GitHub account with the same name as your local project.

1 General

Owner * shamdatta / Repository name * radical-webap
✔ radical-webap is available.

Great repository names are short and memorable. How about [ideal-octo-couscous](#)?

Description
for cidc project
16 / 350 characters

2 Configuration

Choose visibility * Choose who can see and commit to this repository Public

Add README
READMEs can be used as longer descriptions. [About READMEs](#) Off

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#) No .gitignore

Add license
Licenses explain how others can use your code. [About licenses](#) No license

Create repository

GITHUB PUSH

Connect the local Git repository to the GitHub repository and push the project.

```
#git remote add origin https://github.com/<username>/radical-webapp.git
```

```
#git branch -M main
```

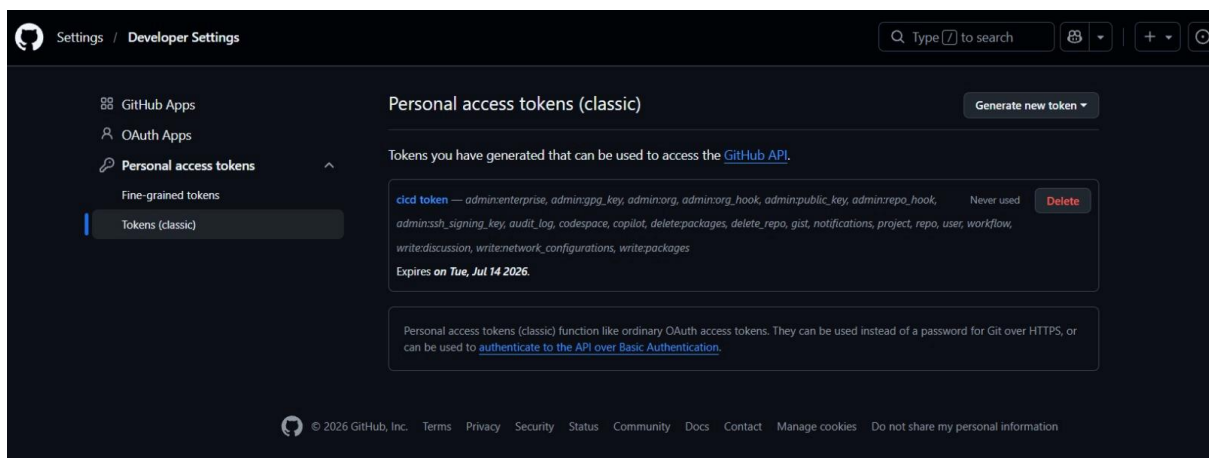
```
#git push -u origin main
```

GitHub will prompt for authentication.

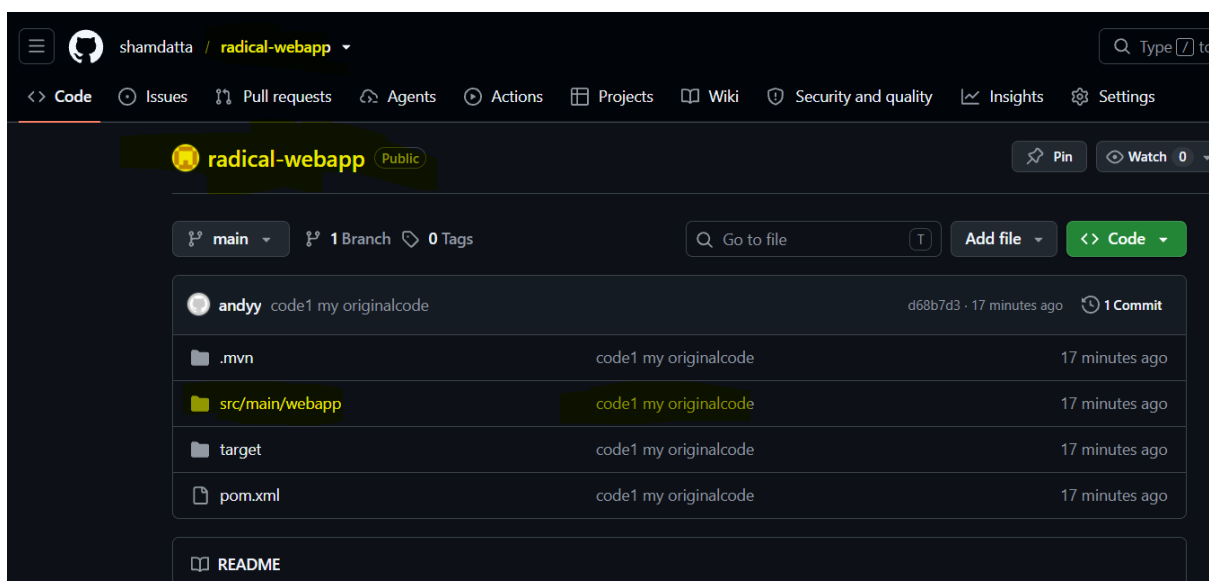
- Enter your GitHub username.
- For the password, use the Personal Access Token (Classic) generated instead of your GitHub account password.

```
[root@jenkins radical-webapp]# git remote add origin https://github.com/shamdatta/radical-webapp.git
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]# git remote -v
origin https://github.com/shamdatta/radical-webapp.git (fetch)
origin https://github.com/shamdatta/radical-webapp.git (push)
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]# git branch -M main
```

```
[root@jenkins radical-webapp]# git push -u origin main
Username for 'https://github.com': andyy
Password for 'https://andyy@github.com':
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 2 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (15/15), 3.23 KiB | 1.62 MiB/s, done.
Total 15 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/shamdatta/radical-webapp.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
[root@jenkins radical-webapp]#
```



Verify that the project has been successfully pushed to GitHub by opening the repository in your browser.



Now make a small change to the project to verify that the changes can be pushed successfully.

```
#vi src/main/webapp/index.jsp
```

```
<html>
<body>
<h2><%= "Hello World! Welcome to our Project!!!"%></h2>
</body>
</html>
```

After making changes, stage the modified files using:

```
# git add .
```

```
# git status
```

```
[root@jenkins radical-webapp]# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   src/main/webapp/index.jsp

no changes added to commit (use "git add" and/or "git commit -a")
[root@jenkins radical-webapp]# git add .
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   src/main/webapp/index.jsp
```

Once the modified file is in the staging area, commit the changes.

```
#git log
```

```
[root@jenkins radical-webapp]# git commit -m "code2 my 2nd code modification"
[main b9cb95b] code2 my 2nd code modification
 1 file changed, 1 insertion(+), 1 deletion(-)
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]# git log
commit b9cb95b9e5db27419863560e439cc6137994bcbd (HEAD -> main)
Author: andyy <andy@example.com>
Date:   Tue Jul 7 21:35:53 2026 +0530

    code2 my 2nd code modification

commit d68b7d32e9b312a16f456ce644f39234e11ac5a6 (origin/main)
Author: andyy <andy@example.com>
Date:   Tue Jul 7 21:12:47 2026 +0530

    code1 my originalcode
[root@jenkins radical-webapp]#
```

Push to GitHub

```
#git remote add origin https://github.com/username/radical-webapp
```

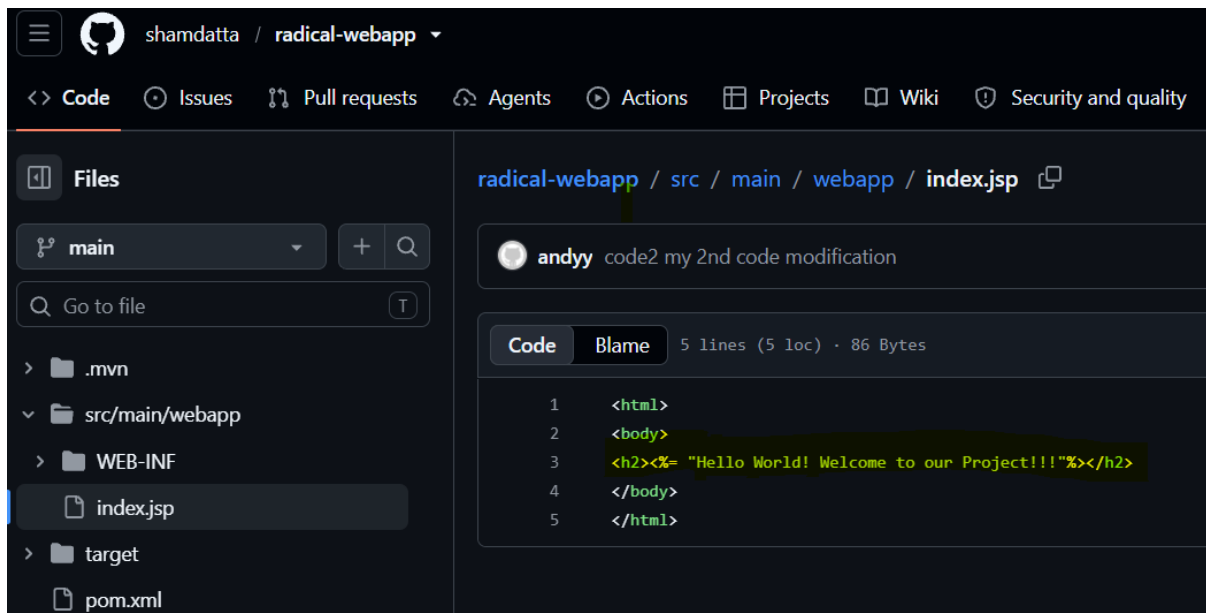
```
#git branch -M main
```

```
#git push -u origin main
```

- Add username and password (the classic token we generated)

```
[root@jenkins radical-webapp]# git push -u origin main
Username for 'https://github.com': andyy
Password for 'https://andyy@github.com':
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 495 bytes | 495.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/shamdatta/radical-webapp.git
    d68b7d3..b9cb95b  main -> main
branch 'main' set up to track 'origin/main'.
[root@jenkins radical-webapp]#
```

Verification -



The screenshot shows the GitHub web interface for the repository 'shamdatta / radical-webapp'. The 'Files' tab is selected, showing the file structure. The file 'index.jsp' is highlighted in the file list on the left. The main content area shows the code for 'index.jsp' with a commit by 'andyy' titled 'code2 my 2nd code modification'. The code is displayed in a dark theme with syntax highlighting. The code content is as follows:

```
1 <html>
2 <body>
3 <h2><%= "Hello World! Welcome to our Project!!!"%></h2>
4 </body>
5 </html>
```

Step 11- Install and configure Jenkins

```
#wget -O /etc/yum.repos.d/jenkins.repo \https://pkg.jenkins.io/rpm-stable/jenkins.repo
```

```
[root@jenkins radical-webapp]# wget -O /etc/yum.repos.d/jenkins.repo \https://pkg.jenkins.io/rpm-stable/jenkins
.repo
--2026-07-07 21:43:49-- https://pkg.jenkins.io/rpm-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.210.133, 2a04:4e42:25::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.210.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.re 100%[=====>]      267  --.-KB/s    in 0s

2026-07-07 21:43:50 (14.9 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]

[root@jenkins radical-webapp]#
```

The Jenkins repository configuration file has been downloaded to /etc/yum.repos.d/.

This allows the yum package manager to install Jenkins from the official Jenkins repository.

Check using –

```
# ll /etc/yum.repos.d
```

```
# yum repolist
```

```
[root@jenkins yum.repos.d]# ll /etc/yum.repos.d/
total 332
-rw-r--r--. 1 root root    101 Jun 25 11:08 AppStream.repo
-rw-r--r--. 1 root root     85 Jun 25 11:08 BaseOS.repo
-rw-r--r--. 1 root root    267 Jun  9 12:44 jenkins.repo
-rw-r--r--. 1 root root 326945 Jul  7 15:17 redhat.repo
[root@jenkins yum.repos.d]#
[root@jenkins yum.repos.d]# yum repolist
Updating Subscription Management repositories.
repo id                                repo name
AppStream                              this is cdrom repo of appstream
BaseOS                                  This is a baseos repo
jenkins                                Jenkins-stable
[root@jenkins yum.repos.d]#
```

Now we will install Jenkins

```
#yum install Jenkins
```

```
[root@jenkins yum.repos.d]# yum install jenkins
Updating Subscription Management repositories.
Jenkins-stable                               1.1 kB/s | 833 B      00:00
Jenkins-stable                               1.6 kB/s | 1.6 kB     00:01
Importing GPG key 0x14ABFC68:
  Userid      : "Jenkins Project <jenkinsci-board@googlegroups.com>"
  Fingerprint: 5E38 6EAD B55F 0150 4CAE 8BCF 7198 F4B7 14AB FC68
  From        : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
Is this ok [y/N]: y
Jenkins-stable                               1.5 kB/s | 2.3 kB     00:01
Last metadata expiration check: 0:00:01 ago on Tue 07 Jul 2026 09:47:26 PM IST.
Dependencies resolved.
=====
Package                        Architecture      Version           Repository        Size
=====
Installing:
jenkins                       noarch            2.555.3-1        jenkins           95 M
=====
Transaction Summary
=====
Install 1 Package

Total download size: 95 M
Installed size: 95 M
Is this ok [y/N]: y
```

Purpose: Install Jenkins on the Linux server so it can automate software build, testing, and deployment tasks.

Enable Jenkins services

```
[root@jenkins yum.repos.d]# systemctl daemon-reload
[root@jenkins yum.repos.d]#
[root@jenkins yum.repos.d]# systemctl enable jenkins
[root@jenkins yum.repos.d]#
[root@jenkins yum.repos.d]# systemctl status jenkins
• jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-07-07 22:23:06 IST; 2min 28s ago
     Main PID: 11866 (java)
        Tasks: 37 (limit: 10324)
      Memory: 456.7M (peak: 463.4M)
         CPU: 29.123s
    CGroup: /system.slice/jenkins.service
            └─11866 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cac>
Jul 07 22:22:54 jenkins.example.com jenkins[11866]: [LF]> This may also be found at: /var/lib/jenkins/secrets/i>
Jul 07 22:22:54 jenkins.example.com jenkins[11866]: [LF]>
Jul 07 22:22:54 jenkins.example.com jenkins[11866]: [LF]> *****
Jul 07 22:22:54 jenkins.example.com jenkins[11866]: [LF]> *****
```

Configure Firewall

Allow TCP port 8080 through the firewall so Jenkins can be accessed from a web browser.

```
#firewall-cmd --permanent--add-port=8080/tcp
```

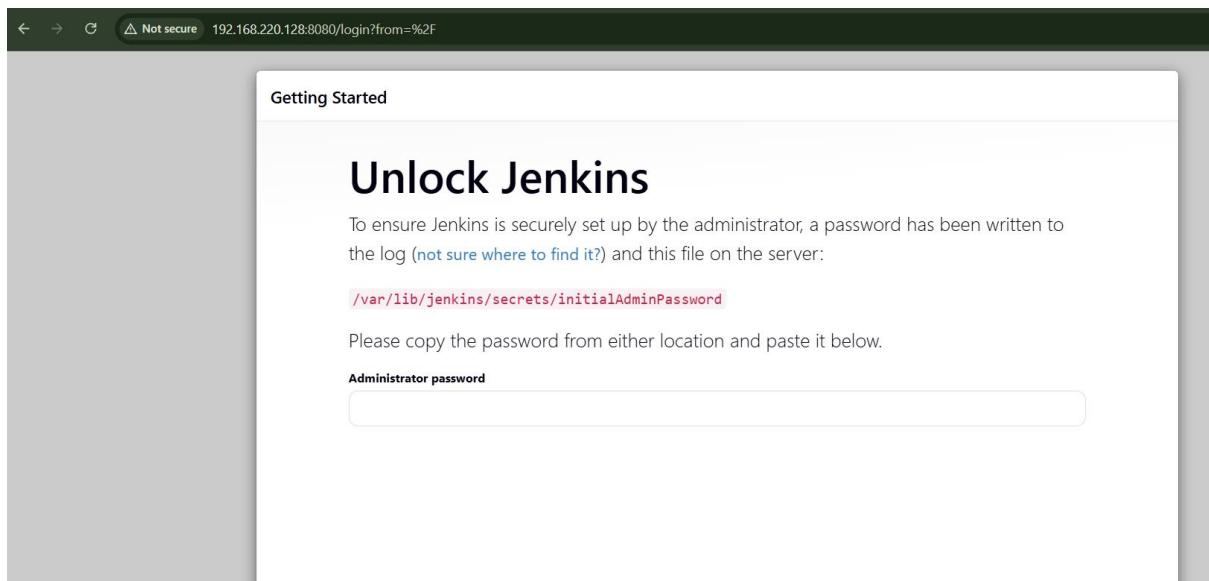
```
#firewall-cmd --reload
```

```
[root@jenkins yum.repos.d]# firewall-cmd --permanent --add-port=8080/tcp
success
[root@jenkins yum.repos.d]# firewall-cmd --reload
success
[root@jenkins yum.repos.d]#
```

Open the following URL in your web browser.

```
#http://192.168.220.126:8080
```

Verification- Jenkins is now accessible through port 8080.



Retrieve the initial administrator password using the following command:

```
#cat /var/lib/jenkins/secrets/initialAdminPassword
```

```
[root@jenkins yum.repos.d]# cat /var/lib/jenkins/secrets/initialAdminPassword
4caeeb9e81b64b0bb9a51b4d1cbd1c26
[root@jenkins yum.repos.d]#
```

Now access Jenkins

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/lib/jenkins/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue

Create the first administrator user for Jenkins.

Getting Started

Create First Admin User

Username

Password

Confirm password

Full name

E-mail address

Jenkins 2.555.3

[Skip and continue as admin](#)

[Save and Continue](#)

Instance Configuration

Jenkins URL:

http://192.168.220.128:8080/

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Jenkins 2.555.3

Not now

Save and Finish

Getting Started

Jenkins is ready!

Your Jenkins setup is complete.

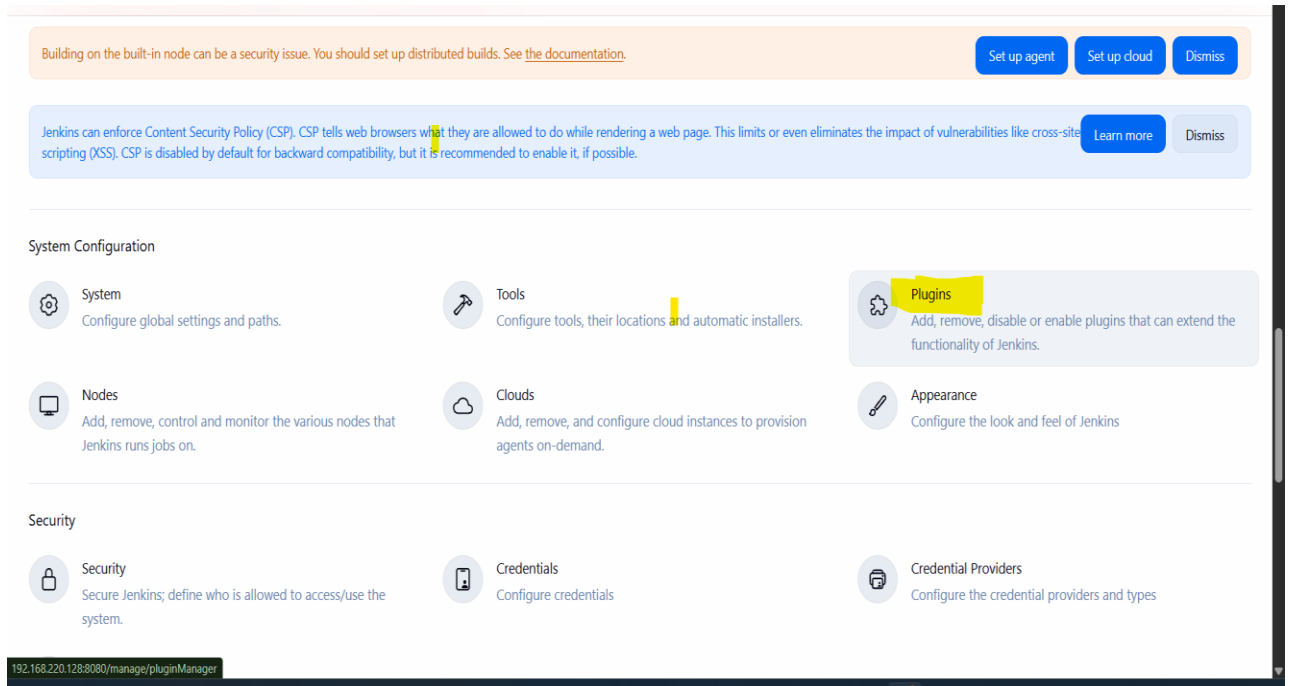
Start using Jenkins

The first administrator account has been created successfully. You can now access the Jenkins dashboard.

Navigate to:

Dashboard → Manage Jenkins → Plugins

Install the following plugins required for the CI/CD pipeline.



Install the following plugins required for the CI/CD pipeline:

- Publish over SSH
- Maven Integration
- GitHub Integration
- Deploy to container

 **Jenkins**

Manage Jenkins

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Search available plugins

Install

Install	Name ↓	Released	Health
☒	Publish Over SSH 390.vb_f56e7405751 Artifact Uploaders Build Tools Send build artifacts over SSH	1 yr 0 mo ago	94
☒	Maven Integration 3.27 Build Tools This plugin provides a deep integration between Jenkins and Maven. It adds support for automatic triggers between projects depending on SNAPSHOTS as well as the automated configuration of various Jenkins publishers such as Junit.	10 mo ago	95
☒	GitHub Integration 0.7.4 emailx Build Triggers GitHub Integration Plugin for Jenkins	1 mo 12 days ago	87
☒	Deploy to container 1.17 Artifact Uploaders This plugin allows you to deploy a war to a container after a successful build. Glassfish 3.x remote deployment	1 yr 1 mo ago	89


Jenkins

[Manage Jenkins](#)

[Plugins](#)

Plugins

- Updates
- Available plugins
- Installed plugins
- Advanced settings
- Download progress**

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

JavaBeans Activation Framework (JAF) API	Success
JAXB	Success
JSON API	Success
Jackson Annotations 2 API	Success
Jakarta Activation API	Success
SnakeYAML API	Success
Jakarta XML Binding API	Success
Woodstox Core API	Success
Jackson 2 API	Success
Infrastructure plugin for Publish Over X	Success
Struts	Success
EDDSA API	Success
Gson API	Success
Trilead API	Success
Variant	Success
commons-lang3 v3.x Jenkins API	Success
Ionicons API	Success
commons-text API	Success

 **Jenkins** / Manage Jenkins ▾ / Plugins

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

bouncycastle API

Credentials

SSH Credentials

JSch dependency

Publish Over SSH

Dev Tools Symbols API

Javadoc

ASM API

SCM API

Caffeine API

Script Security

Pipeline: Step API

Pipeline: API

Pipeline: Supporting APIs

Plugin Utilities API

Font Awesome API

Bootstrap 5 API

SnakeYAML Engine API

Jackson 3 API

jQuery3 API

ECharts API

OWASP Markup Formatter

Prism API

Disclav URI API

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

✓ Success

All the required plugins have been installed successfully.

Run the following command to verify `JAVA_HOME`:

```
#echo $JAVA_HOME
```

```
[root@jenkins plugins]# echo $JAVA_HOME
/usr/lib/jvm/java-21-openjdk
[root@jenkins plugins]#
```

This path will be used while configuring the JDK in Jenkins.

Configure Global Tools

Navigate to:

Dashboard

→ Manage Jenkins

→ Tools

Configure the following tools:

- JDK
- Maven

These tools will be used by Jenkins while building the project.

Maven installations

+ Add Maven

Maven

Name

maven

MAVEN_HOME

/opt/maven

☐ Install automatically ?

JDK installations

+ Add JDK

JDK

Name

java-21

JAVA_HOME

/usr/lib/jvm/java-21-openjdk

☐ Install automatically ?

Create a new Jenkins job named `maven_project` and select **Maven Project** as the project type.

New Item

Enter an item name

`maven_projectt`

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view...

OK

Configure the remaining options as shown in the screenshots below.

Jenkins Job Configuration

Jenkins / `maven_projectt` / Configure

Configure

General

Source Code Management

Triggers

Environment

Pre Steps

Build

Post Steps

Build Settings

Post-build Actions

General

Enabled

Description

Plain text [Preview](#)

☐ Discard old builds [?](#)

☐ GitHub project

☐ This project is parameterized [?](#)

☐ Throttle builds [?](#)

☐ Execute concurrent builds if necessary [?](#)

Advanced

Configure

- General
- Source Code Management
- Triggers
- Environment
- Pre Steps
- Build
- Post Steps
- Build Settings
- Post-build Actions

Git ?

Repositories ?

Repository URL ?

https://github.com/shamdatta/radical-webapp.git

Credentials ?

- none -

+ Add

Advanced ▾

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

main

Configure

- General
- Source Code Management
- Triggers
- Environment
- Pre Steps
- Build
- Post Steps
- Build Settings
- Post-build Actions

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- ☒ Build whenever a SNAPSHOT dependency is built ?
- ☐ Schedule build when some upstream has no successful builds ?
- ☐ Trigger builds remotely (e.g., from scripts) ?
- ☐ Build after other projects are built ?
- ☐ Build periodically ?
- ☐ GitHub Branches
- ☐ GitHub Pull Requests ?
- ☐ GitHub hook trigger for GITScm polling ?

☒ Poll SCM ?

Schedule ?

* * * * *

Configure

- General
- Source Code Management
- Triggers
- Environment
- Pre Steps
- Build
- Post Steps
- Build Settings
- Post-build Actions

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

- ☐ Use secret text(s) or file(s) ?
- ☐ Send files or execute commands over SSH before the build starts ?
- ☐ Send files or execute commands over SSH after the build runs ?

Pre Steps

+ Add pre-build step

Build

Root POM ?

pom.xml

Goals and options ?

clean install

```
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.jar (196 kB at 1.9 MB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.jar (157 kB at 1.9 MB/s)
[INFO] Installing /var/lib/jenkins/workspace/maven_projectt/pom.xml to /var/lib/jenkins/.m2/repository/in/radical/webapp/radical-webapp/1.0-SNAPSHOT/radical-webapp-1.0-SNAPSHOT.pom
[INFO] Installing /var/lib/jenkins/workspace/maven_projectt/target/radical-webapp.war to /var/lib/jenkins/.m2/repository/in/radical/webapp/radical-webapp/1.0-SNAPSHOT/radical-webapp-1.0-SNAPSHOT.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 17.789 s
[INFO] Finished at: 2026-07-07T23:43:36+05:30
[INFO] -----
Waiting for Jenkins to finish collecting data
[JENKINS] Archiving /var/lib/jenkins/workspace/maven_projectt/pom.xml to in.radical.webapp/radical-webapp/1.0-SNAPSHOT/radical-webapp-1.0-SNAPSHOT.pom
[JENKINS] Archiving /var/lib/jenkins/workspace/maven_projectt/target/radical-webapp.war to in.radical.webapp/radical-webapp/1.0-SNAPSHOT/radical-webapp-1.0-SNAPSHOT.war
channel stopped
Finished: SUCCESS
```

The Maven project has now been configured successfully as you can see the console output as success above.

Step 12- Tomcat Server Configuration

Configure Hostname

```
#hostnamectl set-hostname tomcat.example.com
```

```
[root@tomcatsrv1 ~]# hostnamectl set-hostname tomcat.example.com
[root@tomcatsrv1 ~]#
```

```
[root@tomcatsrv1 ~]# ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.220.136 netmask 255.255.255.0 broadcast 192.168.220.255
    inet6 fe80::20c:29ff:fec7:4e76 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:c7:4e:76 txqueuelen 1000 (Ethernet)
    RX packets 175155 bytes 246984807 (235.5 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 28168 bytes 1831892 (1.7 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 22 bytes 2340 (2.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 22 bytes 2340 (2.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```


Install Java

```
#yum install java-21-openjdk -y
```

```
Total                                                                 3.7 MB/s | 50 MB    00:13
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Running scriptlet: java-21-openjdk-headless-1:21.0.11.0.10-2.el9.x86_64 1/1
  Preparing          :                                                                 1/1
  Installing         : java-21-openjdk-headless-1:21.0.11.0.10-2.el9.x86_64 1/2
  Running scriptlet: java-21-openjdk-headless-1:21.0.11.0.10-2.el9.x86_64 1/2
  Installing         : java-21-openjdk-1:21.0.11.0.10-2.el9.x86_64 2/2
  Running scriptlet: java-21-openjdk-1:21.0.11.0.10-2.el9.x86_64 2/2
  Running scriptlet: java-21-openjdk-headless-1:21.0.11.0.10-2.el9.x86_64 2/2
  Running scriptlet: java-21-openjdk-1:21.0.11.0.10-2.el9.x86_64 2/2
  Verifying          : java-21-openjdk-1:21.0.11.0.10-2.el9.x86_64 1/2
  Verifying          : java-21-openjdk-headless-1:21.0.11.0.10-2.el9.x86_64 2/2
Installed products updated.

Installed:
  java-21-openjdk-1:21.0.11.0.10-2.el9.x86_64      java-21-openjdk-headless-1:21.0.11.0.10-2.el9.x86_64

Complete!
[root@tomcatsrv1 ~]#
```

Download Apache Tomcat

```
#cd /opt
```

```
#wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.55/bin/apache-tomcat-10.1.55.tar.gz
```

```
#tar -xvzf apache-tomcat-10.1.55.tar.gz
```

```
#mv apache-tomcat-10.1.55 tomcat
```

```
[root@tomcatsrv1 opt]# wget https://archive.apache.org/dist/tomcat/tomcat-10/v10.1.55/bin/apache-tomcat-10.1.55.tar.gz
--2026-07-07 23:58:55-- https://archive.apache.org/dist/tomcat/tomcat-10/v10.1.55/bin/apache-tomcat-10.1.55.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14343328 (14M) [application/x-gzip]
Saving to: 'apache-tomcat-10.1.55.tar.gz.1'

apache-tomcat-10.1.55.tar.gz.1  100%[=====>] 13.68M  749KB/s  in 19s

2026-07-07 23:59:15 (726 KB/s) - 'apache-tomcat-10.1.55.tar.gz.1' saved [14343328/14343328]

[root@tomcatsrv1 opt]# tar -xvzf apache-tomcat-10.1.55.tar.gz
apache-tomcat-10.1.55/conf/
apache-tomcat-10.1.55/conf/catalina.policy
apache-tomcat-10.1.55/conf/catalina.properties
apache-tomcat-10.1.55/conf/context.xml
apache-tomcat-10.1.55/conf/jaspic-providers.xml
apache-tomcat-10.1.55/conf/jaspic-providers.xsd
apache-tomcat-10.1.55/conf/logging.properties
apache-tomcat-10.1.55/conf/server.xml
apache-tomcat-10.1.55/conf/tomcat-users.xml

[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# mv apache-tomcat-10.1.55 tomcat
[root@tomcatsrv1 opt]# ls
apache-tomcat-10.1.55.tar.gz.1  tomcat
```

Create Tomcat User... For security reasons, Tomcat should run as a dedicated non-root user.

#groupadd --system tomcat

#useradd -r -g tomcat -d /usr/share/tomcat -s /bin/false tomcat

```
[root@tomcatsrv1 opt]# groupadd --system tomcat
groupadd: group 'tomcat' already exists
[root@tomcatsrv1 opt]# useradd -r -g tomcat -d /usr/share/tomcat -s /bin/false tomcat
useradd: user 'tomcat' already exists
```

Assign Ownership

#chown -R tomcat:tomcat /opt/tomcat

```
[root@tomcatsrv1 opt]# chown -R tomcat:tomcat /opt/tomcat
[root@tomcatsrv1 opt]# ll
total 28020
-rw-r--r--. 1 root root 14343328 May  5 19:09 apache-tomcat-10.1.55.tar.gz
-rw-r--r--. 1 root root 14343328 May  5 19:09 apache-tomcat-10.1.55.tar.gz.1
drwxr-xr-x. 10 tomcat tomcat 4096 Jul  8 00:01 tomcat
```

Start Tomcat

cd /opt/tomcat/bin

./startup.sh

```
[root@tomcatsrv1 opt]# cd /opt/tomcat/bin
[root@tomcatsrv1 bin]# ./startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Tomcat started.
[root@tomcatsrv1 bin]#
```

Open Firewall

#firewall-cmd --permanent --add-port=8083/tcp

#firewall-cmd --reload

```
[root@tomcatsrv1 bin]# firewall-cmd --permanent --add-port=8083/tcp
success
[root@tomcatsrv1 bin]# firewall-cmd --reload
success
[root@tomcatsrv1 bin]# netstat -antlpu | grep java
tcp6      0      0 :::8083                :::*                    LISTEN     34210/java
tcp6      0      0 127.0.0.1:8005          :::*                    LISTEN     34210/java
tcp6      0      0 192.168.220.136:8083    192.168.220.1:60208    ESTABLISHED 34210/java
tcp6      0      0 192.168.220.136:8083    192.168.220.1:51731    ESTABLISHED 34210/java
tcp6      0      0 192.168.220.136:8083    192.168.220.1:55124    ESTABLISHED 34210/java
```

Verify

#ss -tulnp | grep 8080

Access from browser

http://<Tomcat-IP>:8080

Not secure 192.168.220.136:8083

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/10.1.55

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Server Status
Manager App
Host Manager

Developer Quick Start

- [Tomcat Setup](#)
- [First Web Application](#)
- [Realms & AAA](#)
- [JDBC DataSources](#)
- [Examples](#)
- [Servlet Specifications](#)
- [Tomcat Versions](#)

Managing Tomcat

For security, access to the `manager.webapp` is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 10.1 access to the manager application is split between different users. [Read more...](#)

[Release Notes](#)
[Changelog](#)
[Migration Guide](#)
[Security Notices](#)

Documentation

[Tomcat 10.1 Documentation](#)
[Tomcat 10.1 Configuration](#)
[Tomcat Wiki](#)

Find additional important configuration information in:

```
$CATALINA_HOME/RUNNING.txt
```

Developers may be interested in:

- [Tomcat 10.1 Bug Database](#)
- [Tomcat 10.1 JavaDocs](#)
- [Tomcat 10.1 Git Repository at GitHub](#)

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).
- [tomcat-users](#)
User support and discussion
- [taglibs-user](#)
User support and discussion for [Apache Taglibs](#)
- [tomcat-dev](#)
Development mailing list, including commit messages

Configure Users

Edit

/opt/tomcat/conf/tomcat-users.xml

Add

```
<role rolename="manager-gui"/>
<role rolename="manager-script"/>
<role rolename="manager-jmx"/>
<role rolename="manager-status"/>
```

```
<user username="admin"
  password="admin"
  roles="manager-gui,manager-script,manager-jmx,manager-status"/>
```

Restart Tomcat.

```
[root@tomcat bin]# vi /opt/tomcat/conf/tomcat-users.xml
[root@tomcat bin]#
[root@tomcat bin]#
[root@tomcat bin]# ./shutdown.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
[root@tomcat bin]# ./startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@tomcat bin]#
```

192.168.220.136:8083/manager/html

Sign in

http://192.168.220.136:8083
Your connection to this site is not private

Username

Password

← → 🔒 Not secure 192.168.220.136:8083/manager/html




Tomcat Web Application Manager

Message:

Manager

[List Applications](#)
[HTML Manager Help](#)
[Manager Help](#)
[Server Status](#)

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/lab6A	None specified	Lab6A	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
					Start Stop Reload Undeploy

Now for tomcat do these changes in our same 'maven_project'

Deployment Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment
- Pre Steps
- Build
- Post Steps
- Build Settings**
- Post-build Actions

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Deploy war/ear to a container

WAR/EAR files ?
target/*.war

Context path ?
/radical-webapp

Containers
Tomcat 9.x Remote

Credentials
admin/*****
+ Add

Tomcat URL ?
http://192.168.220.136:8083

< Add Username with password

×

Scope ?
Global (Jenkins, nodes, items, all child items, etc)

Username
admin

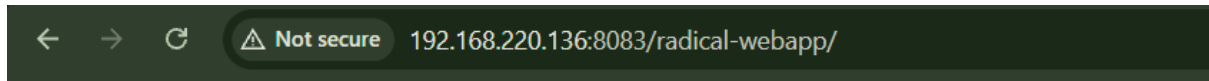
☐ Treat username as secret ?

Password
.....

ID ?
tomcat

Description ?

Create



Hello World! Welcome to our Project!!!

Conclusion

The CI/CD pipeline has been configured successfully.

The pipeline now performs the following workflow:

1. Developer pushes code to GitHub.
2. Jenkins detects the changes.
3. Jenkins builds the Maven project.
4. A WAR file is generated.
5. The application is deployed to Apache Tomcat.
6. Users can access the deployed application through a web browser.